

# AYMANE AIT DADS

Data Science (Cardiac) Intern | Medical Image Analysis · Deep Learning · Computer Vision

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## PROFESSIONAL SUMMARY

Top-10% Kaggle competitor fine-tuning a 30B Mamba-Transformer MoE for multi-type reasoning, demonstrating rigorous large-scale machine learning research with 20+ single-variable ablations on a \$106K+ prize-pool benchmark. Ranked 1st in EURECOM class for AI-generated image detection (0.791 AUC) using CLIP ViT-L/14 fine-tuning on 250K images across 25+ generator types, with submission to the NTIRE 2026 @ CVPR international benchmark. Proficient in PyTorch, deep learning, computer vision, medical-grade prototype validation, and ablation-driven research methodology. Available for internship immediately, based in Sophia Antipolis, France, with full EU work authorization and no visa requirement.

## CORE COMPETENCIES

Medical Image Analysis | Deep Learning | Machine Learning | Prototype Validation | Computer Vision | Algorithm Research | Large-Scale Data Analysis | Model Evaluation

## WORK EXPERIENCE

### Orange Maroc

Jul 2024 - Aug 2024

**Data Science Intern** - Casablanca, Morocco

- Built and evaluated Random Forest classification and K-Means clustering models on **100K+ fiber-optic network records**, applying rigorous machine learning research methodology to a large-scale industrial dataset.
- Developed an end-to-end clustering pipeline that mapped customer performance profiles to **5 commercial service tiers**, output adopted directly by the network optimization team for maintenance decisions.
- Reduced manual analysis time by **~60%** through automated feature engineering, model evaluation workflows, and systematic EDA on upload speed, latency, and jitter signals.
- Delivered stakeholder presentations translating complex model outputs into actionable maintenance recommendations, bridging data science research and **real-world prototype validation**.

## PROJECTS

**AI-Generated Image Detection - EURECOM ImSecu + EURECOM Course + International Benchmark · 2025 · 1st in Class Leaderboard NTIRE 2026 @ CVPR**

- Ranked **1st in EURECOM class private Kaggle leaderboard (0.791 AUC)** for binary image classification using CLIP ViT-L/14 (428M params) fine-tuned on 250K images across 25+ generator types; separately submitted to the NTIRE 2026 @ CVPR CodaBench international benchmark.
- Designed ablation studies across unfreezing depth (4/6/8 transformer blocks), training epochs (1 vs. 4), and ensemble weighting, confirming 1-epoch training generalizes better out-of-distribution, with 10-view TTA and dual learning-rate scheduling for deep learning prototype validation.

PyTorch · OpenCLIP · CLIP ViT-L/14 · Kaggle · FP16 Mixed Precision

**NVIDIA Nemotron Reasoning Challenge - Kaggle Competition** Kaggle · June 2026 · Top 10% Public Leaderboard · \$106K+ Prize Pool (In Progress)

- Achieved **top 10% public leaderboard score (0.80/1.0)** on a \$106K+ prize-pool benchmark by fine-tuning a 30B Mamba-Transformer MoE model with LoRA (r=32, ~888M trainable parameters) on 7,828 chain-of-thought reasoning examples across 6 logic puzzle types.
- Resolved 5 Blackwell GPU (sm\_120) incompatibilities in strict dependency order - including monkey-patched CUDA allocators, stubbed Mamba imports, and PyTorch fallback kernels - and conducted **20+ single-variable ablations** (packing, loss masking, adapter scale) to isolate a 0.80 vs. 0.64 score gap.

PyTorch · Unsloth · PEFT · Hugging Face Transformers · vLLM · Triton

### Anomalous Sound Detection - Industrial Equipment

EURECOM · 2025 · AUC 0.80

- Designed an unsupervised transformer autoencoder on **Mel spectrograms with SpecAugment** for fault detection on slide-rail machinery with no labeled anomaly examples, achieving AUC 0.80.
- Applied deep learning research methodology for prototype validation of a predictive maintenance algorithm, directly analogous to **clinical signal anomaly detection** in medical image analysis pipelines.

PyTorch · Transformer Autoencoder · SpecAugment · Mel Spectrograms

## EDUCATION

**Engineering Degree in Data Science (Master-Level)** - EURECOM

Sep 2023 - Present

Relevant coursework: Machine Learning, Deep Learning, Computer Vision, Image Security, NLP, Statistics, Cloud Computing

**CPGE - Mathematics & Physics** - Ibn Timiya

2021 - 2023

SKILLS

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**Deep Learning & Computer Vision:** PyTorch | CLIP ViT-L/14 | CNN | DenseNet | TensorFlow | FP16 Mixed Precision | Image Classification

**ML Research & LLM Fine-Tuning:** LoRA / QLoRA | PEFT | SFTTrainer | Hugging Face Transformers | Scikit-learn | Ablation Studies | Ensemble Methods

**Data Science & MLOps:** Python | Pandas | NumPy | SQL | Git | Kaggle GPU | Power BI | Feature Engineering